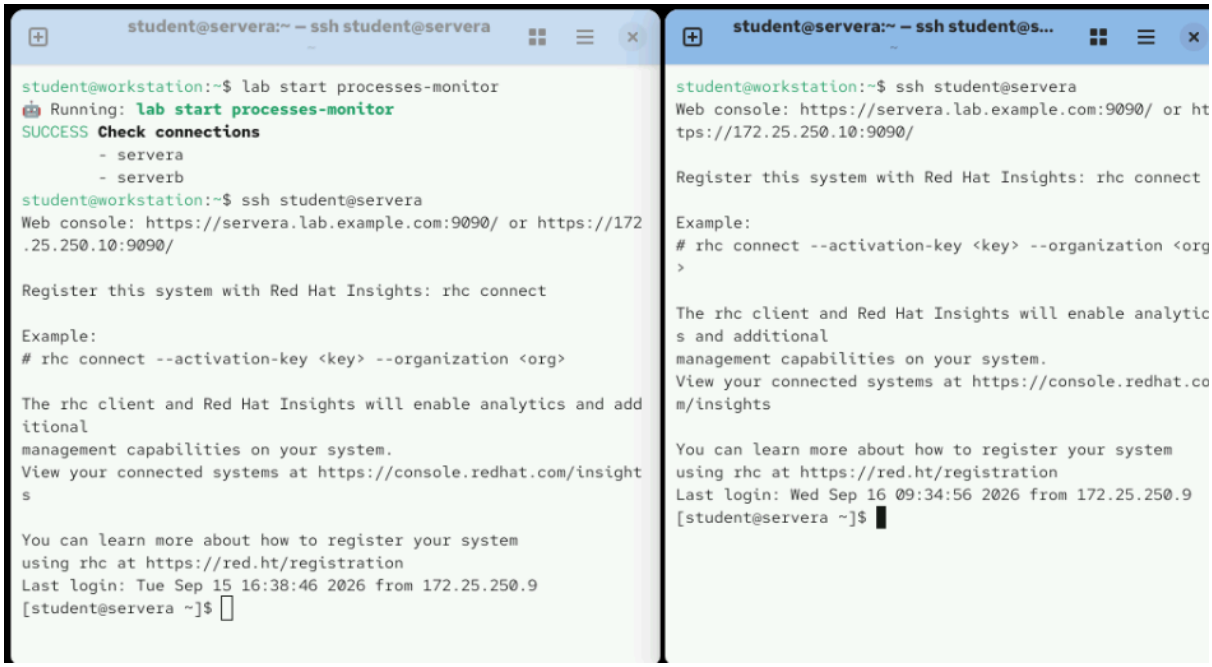
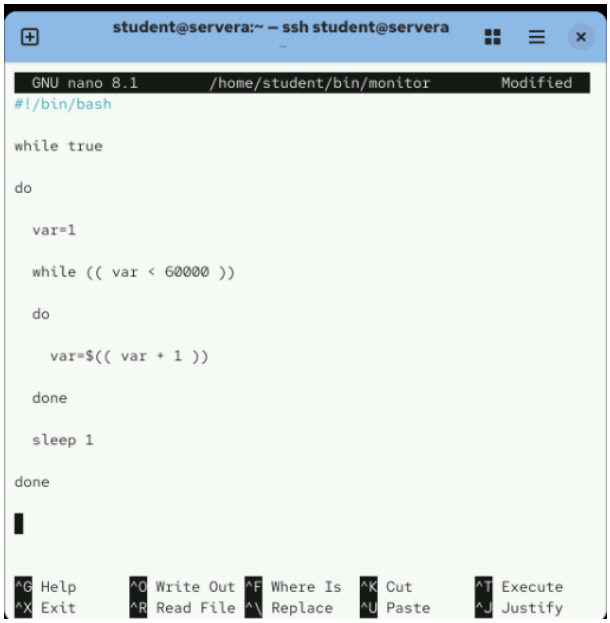
15.8-Guided Exercise: Monitor Process Activity

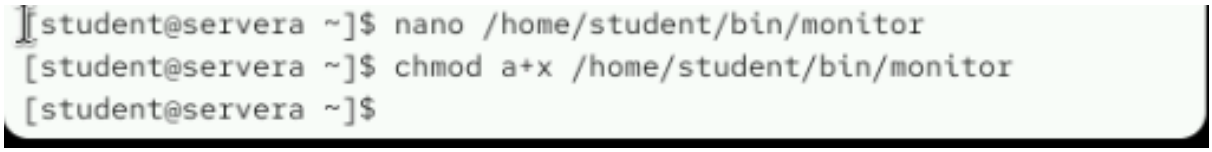
Ahmed Abdelhedi

1/

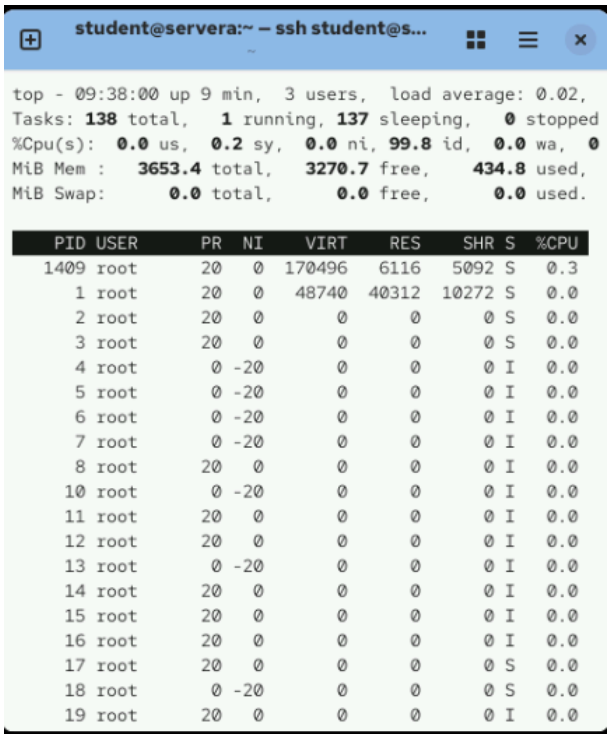


2/





3/

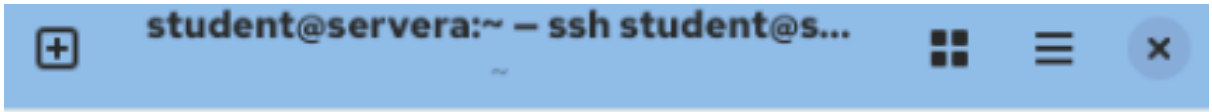

student@servera:~ – ssh student@s...
top - 09:38:00 up 9 min, 3 users, load average: 0.02,
Tasks: 138 total, 1 running, 137 sleeping, 0 stopped
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0
MiB Mem : 3653.4 total, 3270.7 free, 434.8 used,
MiB Swap: 0.0 total, 0.0 free, 0.0 used.
PID USER PR NI VIRT RES SHR S %CPU
1409 root 20 0 170496 6116 5092 S 0.3
1 root 20 0 48740 40312 10272 S 0.0
2 root 20 0 0 0 0 S 0.0
3 root 20 0 0 0 0 S 0.0
4 root 0 -20 0 0 0 I 0.0
5 root 0 -20 0 0 0 I 0.0
6 root 0 -20 0 0 0 I 0.0
7 root 0 -20 0 0 0 I 0.0
8 root 20 0 0 0 0 I 0.0
10 root 0 -20 0 0 0 I 0.0
11 root 20 0 0 0 0 I 0.0
12 root 20 0 0 0 0 I 0.0
13 root 0 -20 0 0 0 I 0.0
14 root 20 0 0 0 0 I 0.0
15 root 20 0 0 0 0 I 0.0
16 root 20 0 0 0 0 I 0.0
17 root 20 0 0 0 0 S 0.0
18 root 0 -20 0 0 0 S 0.0
19 root 20 0 0 0 0 I 0.0

4/


student@servera:~ – ssh student@servera
[student@servera ~]$ nano /home/student/bin/monitor
[student@servera ~]$ chmod a+x /home/student/bin/monitor
[student@servera ~]$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 46 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: Intel(R) Xeon(R) Platinum 8168 CPU @ 2.70GHz
CPU family: 6
Model: 85
Thread(s) per core: 1
Core(s) per socket: 1
Socket(s): 2
Stepping: 4
BogoMIPS: 5387.34
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe po

5/

```
[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$
```




```
top - 09:40:50 up 12 min,  3 users,  load average: 0.11,
Tasks: 141 total,   2 running, 139 sleeping,   0 stopped
%Cpu(s):  6.5 us,  2.2 sy,  0.0 ni, 90.7 id,  0.2 wa,  0
MiB Mem :  11.9/3653.4   [||||                          ]
MiB Swap:   0.0/0.0      [                              ]

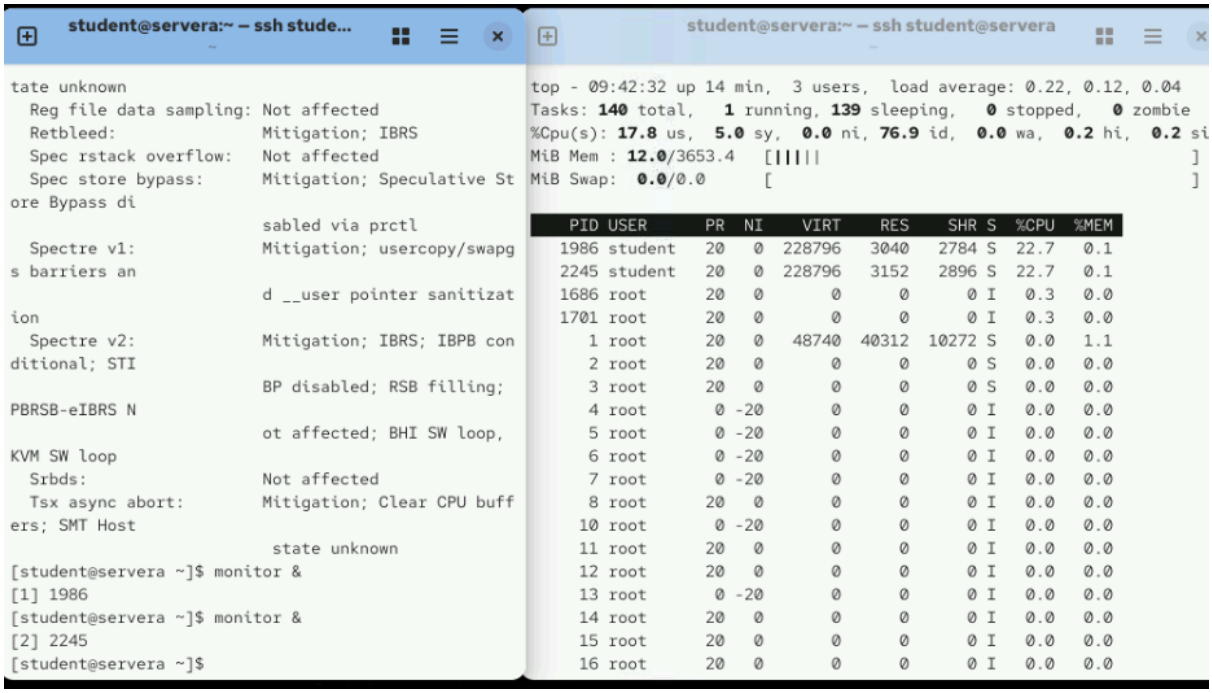
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU
   1986 student   20   0  228796   3040   2784 S  15.9
   1294 cockpit+  20   0   82664   5152   4512 S   0.7
     45 root      39  19       0      0      0 S   0.3
    948 root      20   0   80408   3524   3268 S   0.3
   1763 student   20   0   14900   7152   5256 S   0.3
   1910 student   20   0  232688   5684   3508 R   0.3
      1 root      20   0   48740  40312  10272 R   0.0
      2 root      20   0       0      0      0 S   0.0
      3 root      20   0       0      0      0 S   0.0
      4 root       0 -20       0      0      0 I   0.0
      5 root       0 -20       0      0      0 I   0.0
      6 root       0 -20       0      0      0 I   0.0
      7 root       0 -20       0      0      0 I   0.0
      8 root      20   0       0      0      0 I   0.0
     10 root       0 -20       0      0      0 I   0.0
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1986	student	20	0	228796	3040	2784	S	16.6	0.1
34	root	20	0	0	0	0	I	0.3	0.0
1294	cockpit+	20	0	82664	5152	4512	S	0.3	0.1
1910	**student**	**20**	**0**	**232688**	**5684**	**3508**	**R**	**0.3**	**0.2**
1	root	20	0	48740	40312	10272	S	0.0	1.1
2	root	20	0	0	0	0	S	0.0	0.0
3	root	20	0	0	0	0	S	0.0	0.0
4	root	0	-20	0	0	0	I	0.0	0.0
5	root	0	-20	0	0	0	I	0.0	0.0

7/

```
[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$
```

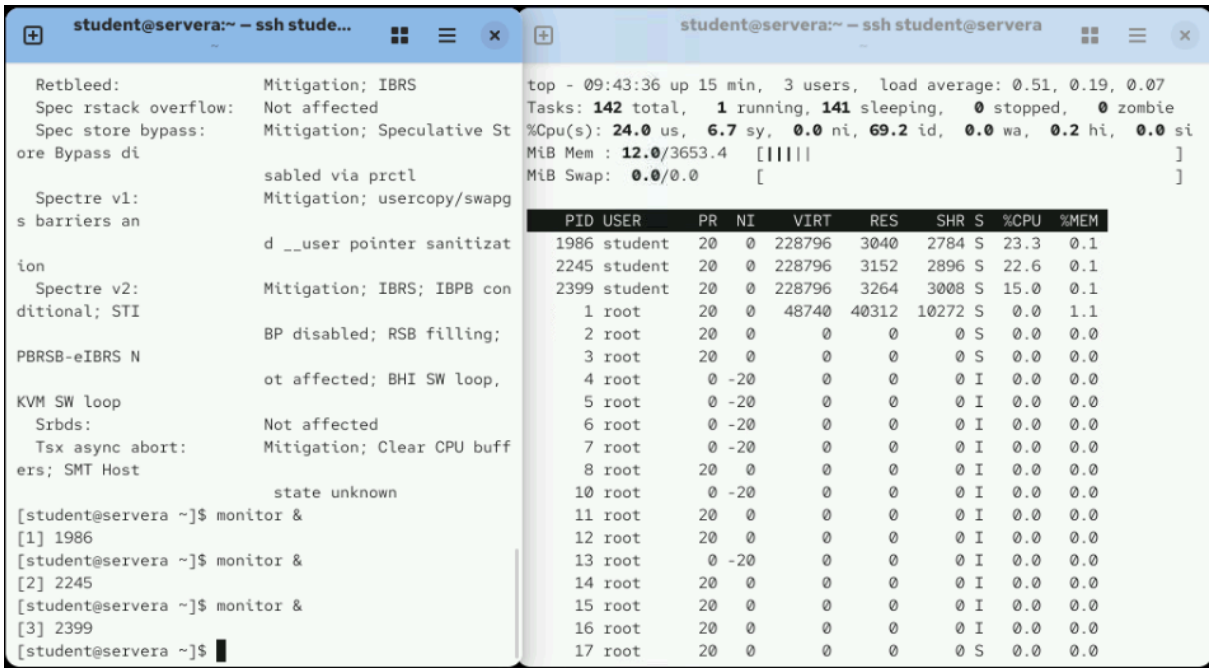
8/


student@servera:~ – ssh stude...
tate unknown
Reg file data sampling: Not affected
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative St
ore Bypass di
sabled via prctl
Spectre v1: Mitigation; usercopy/swapg
s barriers an
d __user pointer sanitizat
ion
Spectre v2: Mitigation; IBRS; IBPB con
ditional; STI
BP disabled; RSB filling;
PBRSB-eIBRS N
ot affected; BHI SW loop,
KVM SW loop
Srbds: Not affected
Tsx async abort: Mitigation; Clear CPU buff
ers; SMT Host
state unknown
[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$
student@servera:~ – ssh student@servera
top - 09:42:32 up 14 min, 3 users, load average: 0.22, 0.12, 0.04
Tasks: 140 total, 1 running, 139 sleeping, 0 stopped, 0 zombie
%Cpu(s): 17.8 us, 5.0 sy, 0.0 ni, 76.9 id, 0.0 wa, 0.2 hi, 0.2 si
MiB Mem : 12.0/3653.4
MiB Swap: 0.0/0.0
PID USER PR NI VIRT RES SHR S %CPU %MEM
1986 student 20 0 228796 3040 2784 S 22.7 0.1
2245 student 20 0 228796 3152 2896 S 22.7 0.1
1686 root 20 0 0 0 0 I 0.3 0.0
1701 root 20 0 0 0 0 I 0.3 0.0
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
8 root 20 0 0 0 0 I 0.0 0.0
10 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0

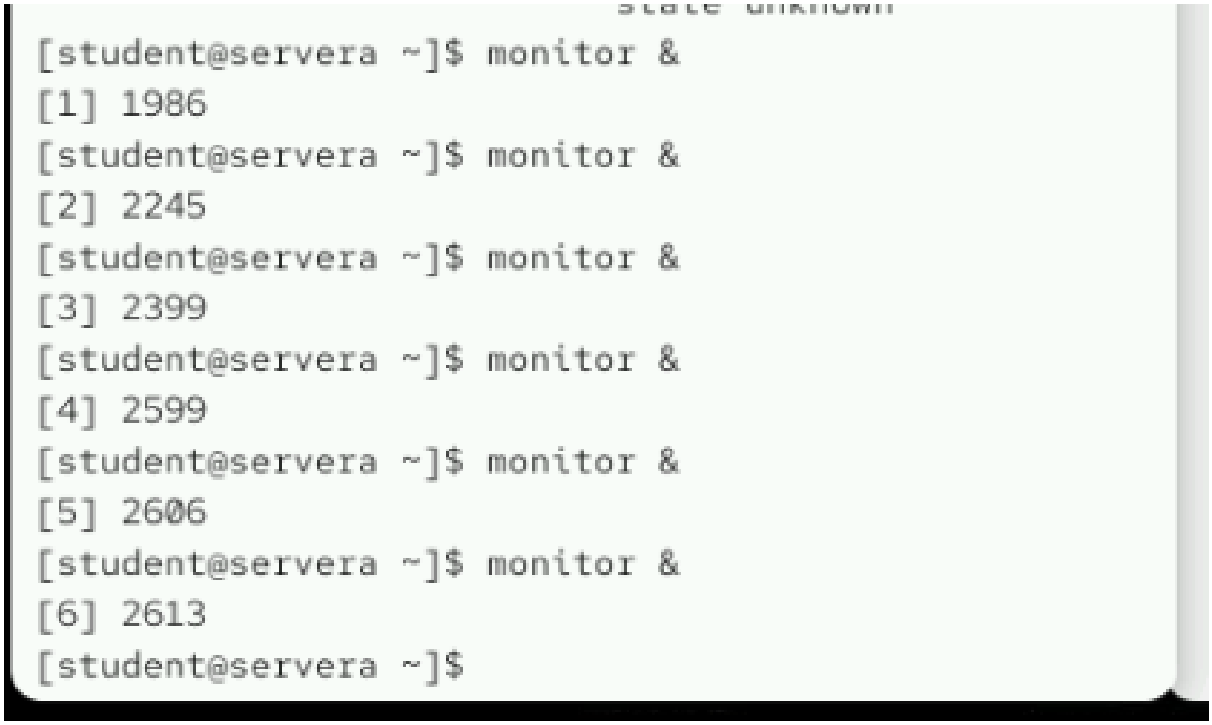
9/


[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$ monitor &
[3] 2399
[student@servera ~]$

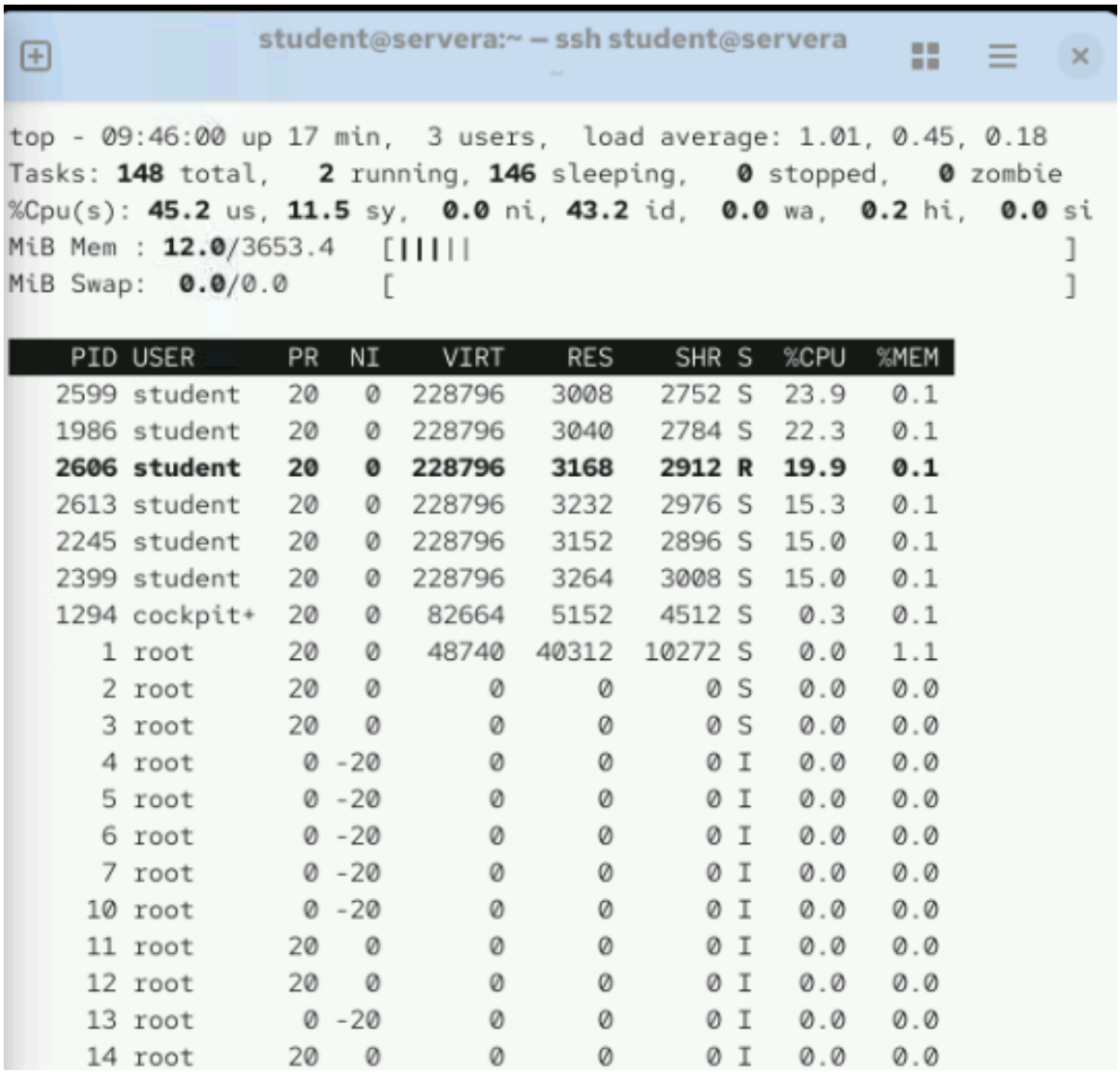
10/


student@servera:~ – ssh stude...
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; IBRS; IBPB conditional; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI SW loop, KVM SW loop
Srbds: Not affected
Tsx async abort: Mitigation; Clear CPU buffers; SMT Host state unknown
[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$ monitor &
[3] 2399
[student@servera ~]$
student@servera:~ – ssh student@servera
top - 09:43:36 up 15 min, 3 users, load average: 0.51, 0.19, 0.07
Tasks: 142 total, 1 running, 141 sleeping, 0 stopped, 0 zombie
%Cpu(s): 24.0 us, 6.7 sy, 0.0 ni, 69.2 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 12.0/3653.4
MiB Swap: 0.0/0.0
PID USER PR NI VIRT RES SHR S %CPU %MEM
1986 student 20 0 228796 3040 2784 S 23.3 0.1
2245 student 20 0 228796 3152 2896 S 22.6 0.1
2399 student 20 0 228796 3264 3008 S 15.0 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
8 root 20 0 0 0 0 I 0.0 0.0
10 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0

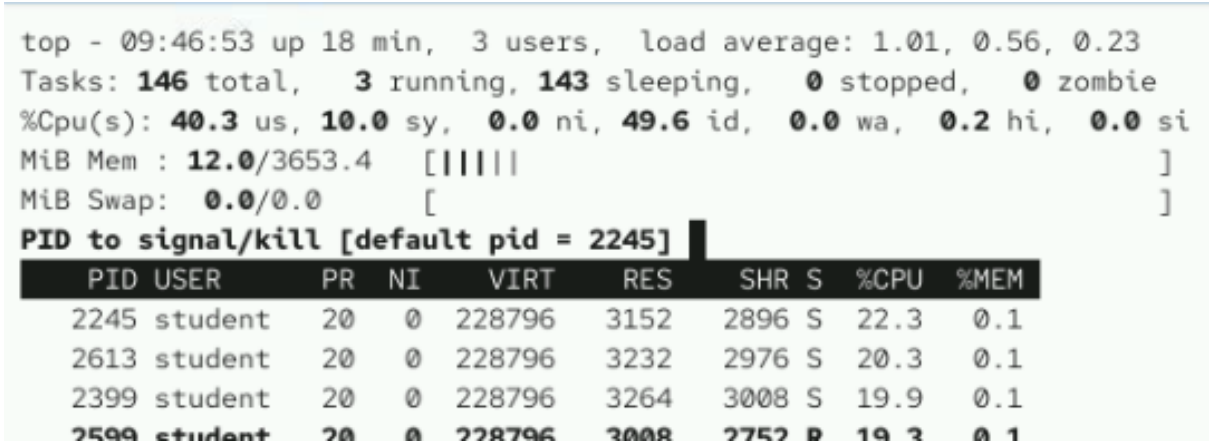
11/

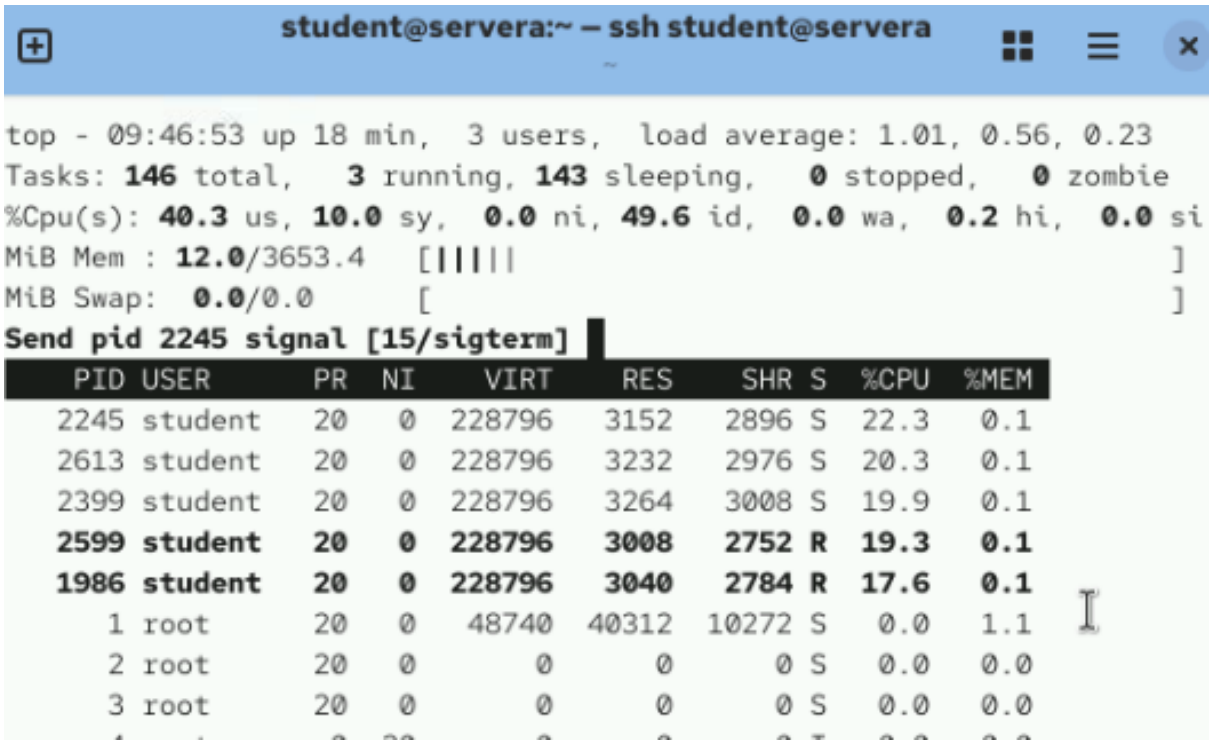

[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$ monitor &
[3] 2399
[student@servera ~]$ monitor &
[4] 2599
[student@servera ~]$ monitor &
[5] 2606
[student@servera ~]$ monitor &
[6] 2613
[student@servera ~]$

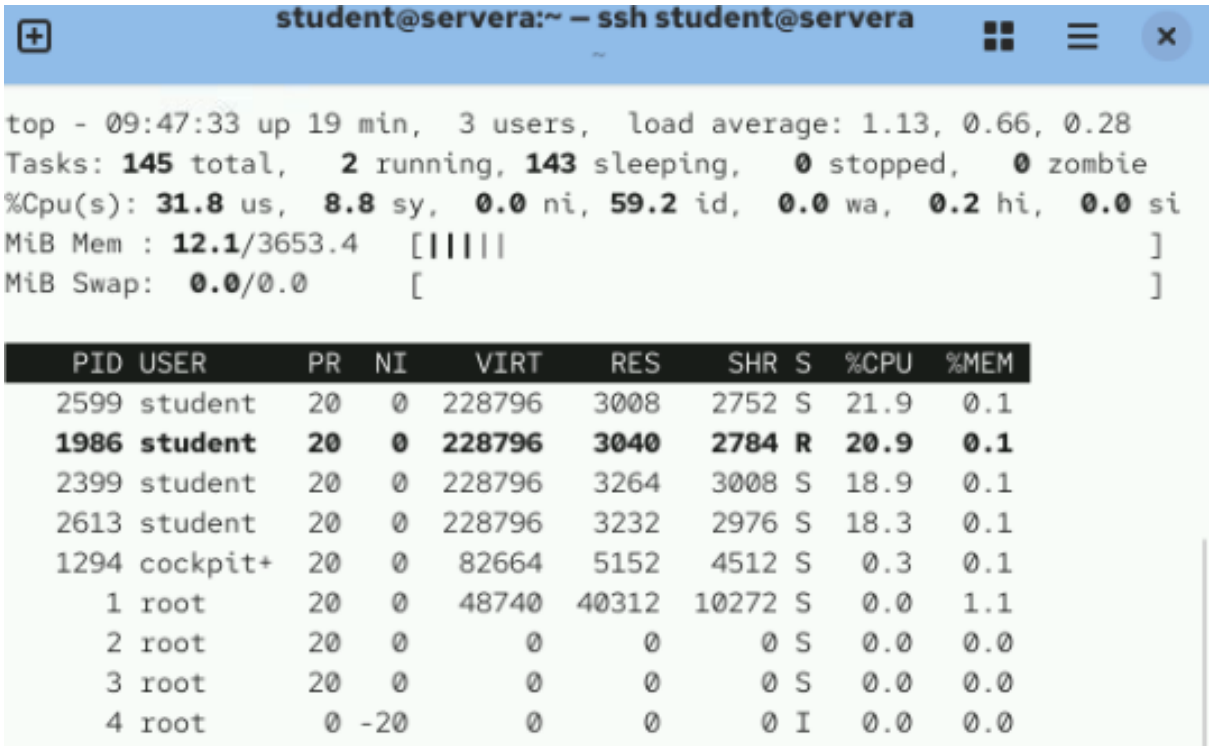
12/


student@servera:~ — ssh student@servera
top - 09:46:00 up 17 min, 3 users, load average: 1.01, 0.45, 0.18
Tasks: 148 total, 2 running, 146 sleeping, 0 stopped, 0 zombie
%Cpu(s): 45.2 us, 11.5 sy, 0.0 ni, 43.2 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 12.0/3653.4 [|||||
MiB Swap: 0.0/0.0 [
PID USER PR NI VIRT RES SHR S %CPU %MEM
2599 student 20 0 228796 3008 2752 S 23.9 0.1
1986 student 20 0 228796 3040 2784 S 22.3 0.1
2606 student 20 0 228796 3168 2912 R 19.9 0.1
2613 student 20 0 228796 3232 2976 S 15.3 0.1
2245 student 20 0 228796 3152 2896 S 15.0 0.1
2399 student 20 0 228796 3264 3008 S 15.0 0.1
1294 cockpit+ 20 0 82664 5152 4512 S 0.3 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
10 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0

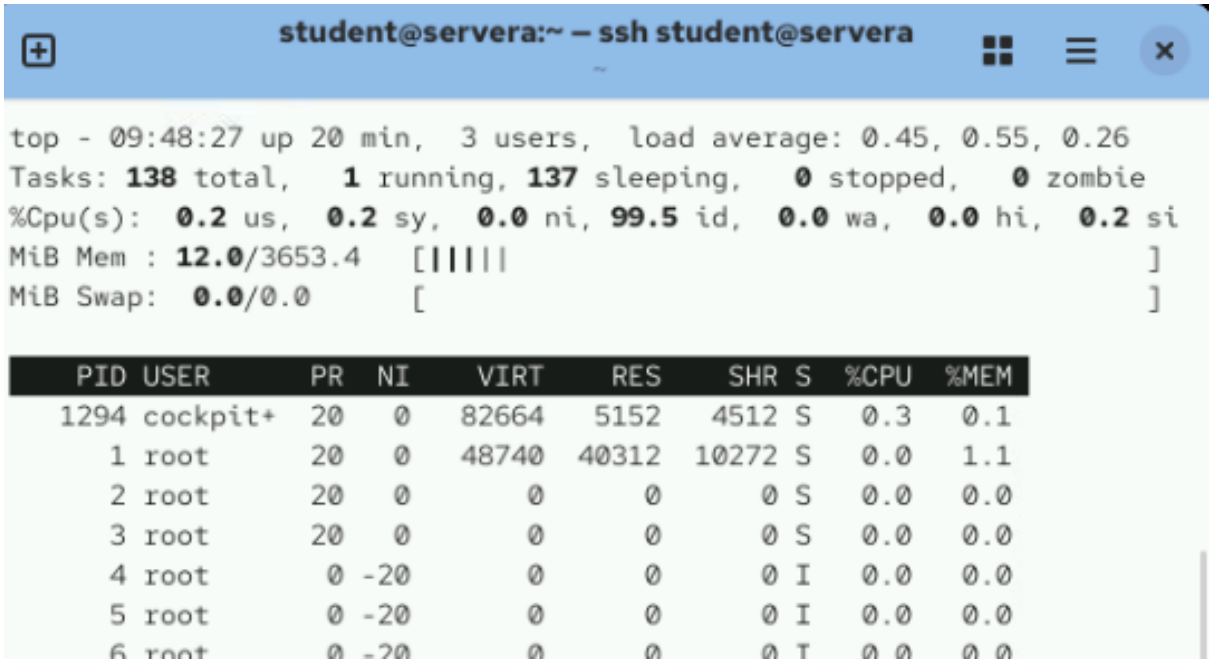
13/


top - 09:46:53 up 18 min, 3 users, load average: 1.01, 0.56, 0.23
Tasks: 146 total, 3 running, 143 sleeping, 0 stopped, 0 zombie
%Cpu(s): 40.3 us, 10.0 sy, 0.0 ni, 49.6 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 12.0/3653.4 [|||||
MiB Swap: 0.0/0.0 [
PID to signal/kill [default pid = 2245]
PID USER PR NI VIRT RES SHR S %CPU %MEM
2245 student 20 0 228796 3152 2896 S 22.3 0.1
2613 student 20 0 228796 3232 2976 S 20.3 0.1
2399 student 20 0 228796 3264 3008 S 19.9 0.1
2599 student 20 0 228796 3008 2752 R 19.3 0.1

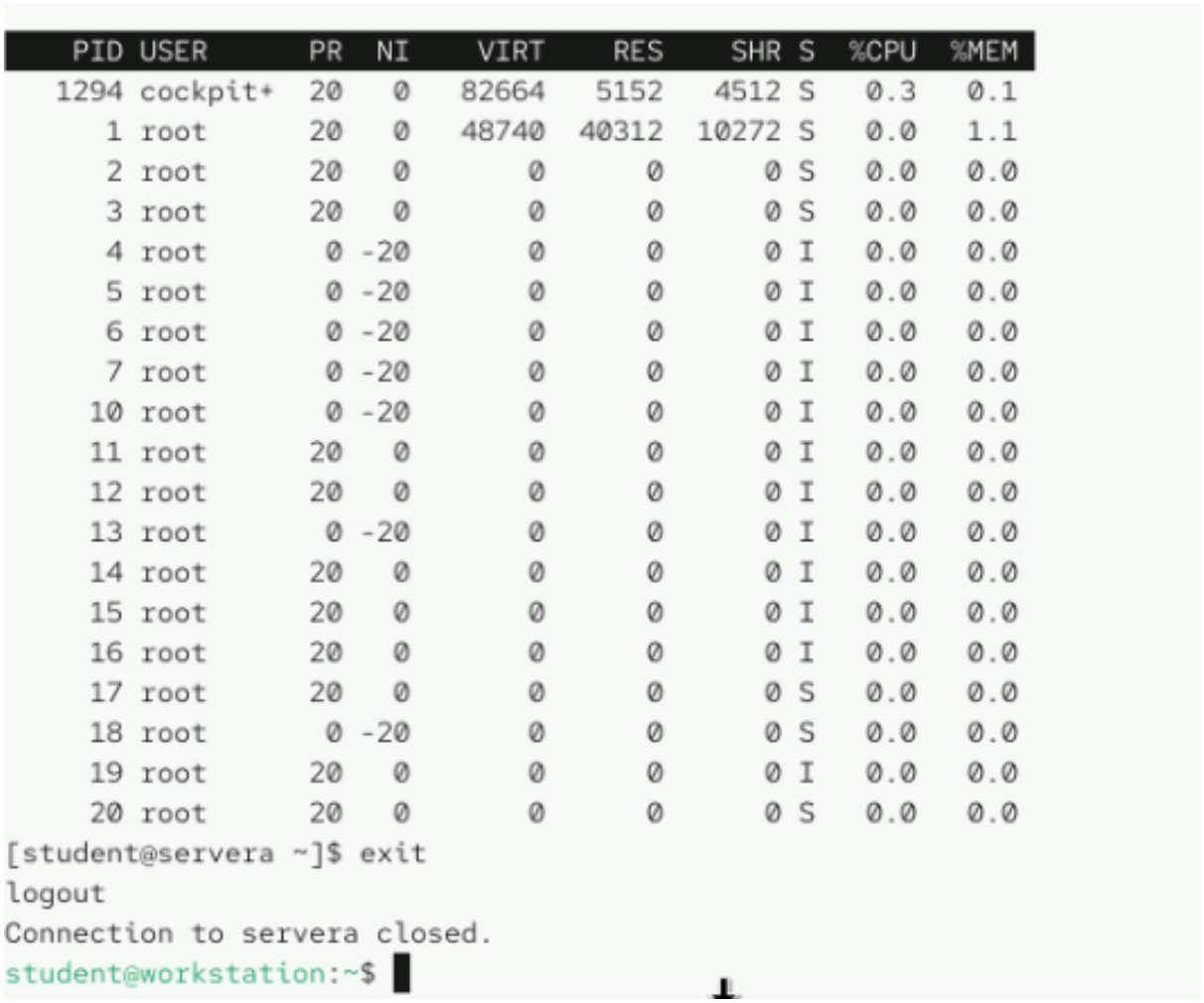

student@servera:~ – ssh student@servera
top - 09:46:53 up 18 min, 3 users, load average: 1.01, 0.56, 0.23
Tasks: 146 total, 3 running, 143 sleeping, 0 stopped, 0 zombie
%Cpu(s): 40.3 us, 10.0 sy, 0.0 ni, 49.6 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 12.0/3653.4 [|||||
MiB Swap: 0.0/0.0 [
Send pid 2245 signal [15/sigterm]
PID USER PR NI VIRT RES SHR S %CPU %MEM
2245 student 20 0 228796 3152 2896 S 22.3 0.1
2613 student 20 0 228796 3232 2976 S 20.3 0.1
2399 student 20 0 228796 3264 3008 S 19.9 0.1
2599 student 20 0 228796 3008 2752 R 19.3 0.1
1986 student 20 0 228796 3040 2784 R 17.6 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0


student@servera:~ – ssh student@servera
top - 09:47:33 up 19 min, 3 users, load average: 1.13, 0.66, 0.28
Tasks: 145 total, 2 running, 143 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.8 us, 8.8 sy, 0.0 ni, 59.2 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 12.1/3653.4 [|||||
MiB Swap: 0.0/0.0 [
PID USER PR NI VIRT RES SHR S %CPU %MEM
2599 student 20 0 228796 3008 2752 S 21.9 0.1
1986 student 20 0 228796 3040 2784 R 20.9 0.1
2399 student 20 0 228796 3264 3008 S 18.9 0.1
2613 student 20 0 228796 3232 2976 S 18.3 0.1
1294 cockpit+ 20 0 82664 5152 4512 S 0.3 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0

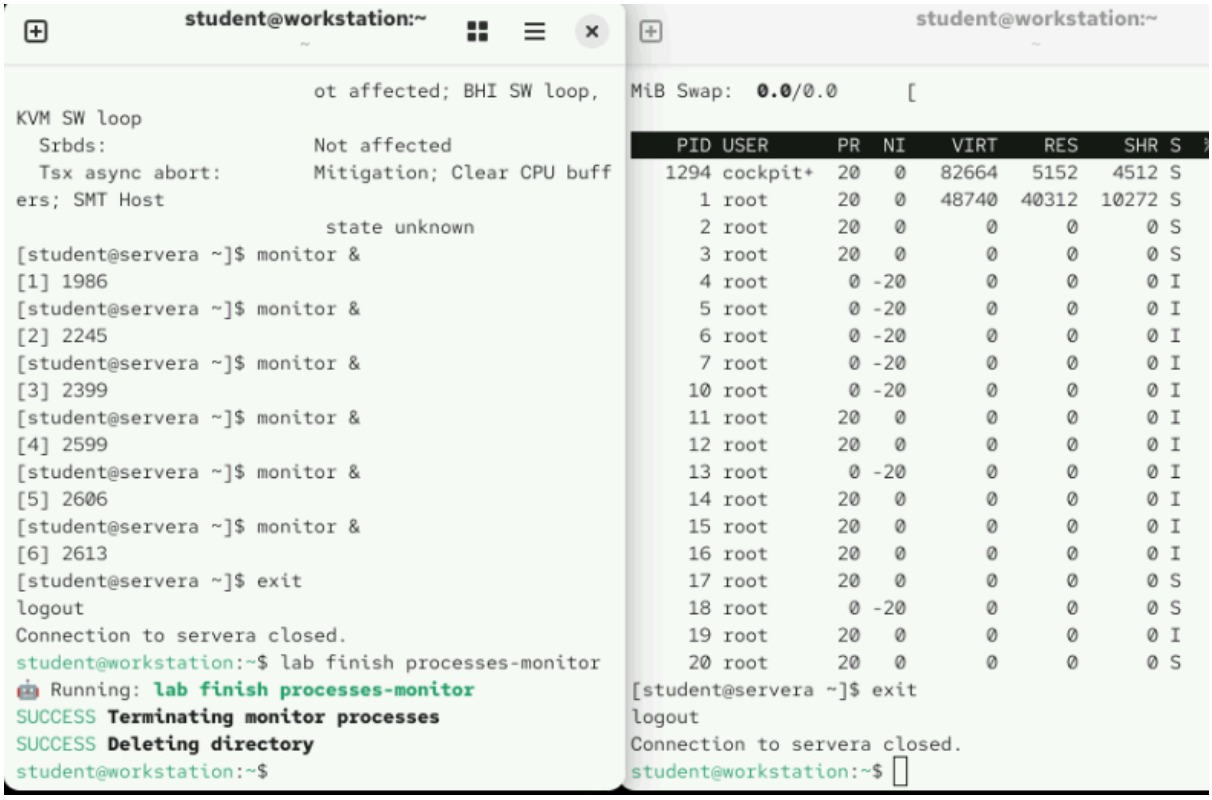
14/


student@servera:~ – ssh student@servera
top - 09:48:27 up 20 min, 3 users, load average: 0.45, 0.55, 0.26
Tasks: 138 total, 1 running, 137 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.2 si
MiB Mem : 12.0/3653.4 [|||||
MiB Swap: 0.0/0.0 [
PID USER PR NI VIRT RES SHR S %CPU %MEM
1294 cockpit+ 20 0 82664 5152 4512 S 0.3 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0

15/


PID USER PR NI VIRT RES SHR S %CPU %MEM
1294 cockpit+ 20 0 82664 5152 4512 S 0.3 0.1
1 root 20 0 48740 40312 10272 S 0.0 1.1
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
10 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0
18 root 0 -20 0 0 0 S 0.0 0.0
19 root 20 0 0 0 0 I 0.0 0.0
20 root 20 0 0 0 0 S 0.0 0.0
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$

16/


student@workstation:~
ot affected; BHI SW loop, KVM SW loop
Srbds: Not affected
Tsx async abort: Mitigation; Clear CPU buffers; SMT Host state unknown
[student@servera ~]$ monitor &
[1] 1986
[student@servera ~]$ monitor &
[2] 2245
[student@servera ~]$ monitor &
[3] 2399
[student@servera ~]$ monitor &
[4] 2599
[student@servera ~]$ monitor &
[5] 2606
[student@servera ~]$ monitor &
[6] 2613
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$ lab finish processes-monitor
Running: lab finish processes-monitor
SUCCESS Terminating monitor processes
SUCCESS Deleting directory
student@workstation:~$
student@workstation:~
MiB Swap: 0.0/0.0
PID USER PR NI VIRT RES SHR S
1294 cockpit+ 20 0 82664 5152 4512 S
1 root 20 0 48740 40312 10272 S
2 root 20 0 0 0 0 S
3 root 20 0 0 0 0 S
4 root 0 -20 0 0 0 I
5 root 0 -20 0 0 0 I
6 root 0 -20 0 0 0 I
7 root 0 -20 0 0 0 I
10 root 0 -20 0 0 0 I
11 root 20 0 0 0 0 I
12 root 20 0 0 0 0 I
13 root 0 -20 0 0 0 I
14 root 20 0 0 0 0 I
15 root 20 0 0 0 0 I
16 root 20 0 0 0 0 I
17 root 20 0 0 0 0 S
18 root 0 -20 0 0 0 S
19 root 20 0 0 0 0 I
20 root 20 0 0 0 0 S
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$